An Explainable Spatial–Frequency Multiscale Transformer for Remote Sensing Scene Classification

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***Abstract*— Deep convolutional neural networks (CNNs) are significant in remote sensing. Due to the strong local representation learning ability, CNNs have excellent performance in remote sensing scene classification. However, CNNs focus on location-sensitive representations in the spatial domain and lack contextual information mining capabilities. Meanwhile, remote sensing scene classification still faces challenges, such as complex scenes and significant differences in target sizes. To address the problems and challenges above, more robust feature representation learning networks are necessary. In this article, a novel and explainable spatial–frequency multiscale Transformer framework, SF-MSFormer, is proposed for remote sensing scene classification. It mainly comprises spatial-domain and frequency-domain multiscale Transformer branches, which consider the spatial–frequency global multiscale representation features. Besides, the texture-enhanced encoder is designed in the frequency-domain multiscale Transformer branch, which is adaptive to capture the global texture features. In addition, an adaptive feature aggregation module is designed to integrate the spatial–frequency multiscale feature for final recognition. The experimental results verify the effectiveness of SF-MSFormer and show better convergence. It achieves state-of-the-art results [98.72%, 98.6%, 99.72%, and 94.83% overall accuracies (OAs), respectively] on the AID, UCM, WHU-RS19, and NWPU-RESISC45 datasets. Besides, the feature visualizations evaluate the explainability of the texture-enhanced encoder. The code implementation of this article will be available at https://github.com/yutinyang/SF-MSFormer.**

***Index Terms*— Convolutional neural networks (CNNs), multiscale representation, spatial–frequency, texture-enhanced encoder, Transformer.**

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I. Introduction

REMOTE sensing scene interpretation has important guidance and research significance for the military, agriculture, etc. [1]. In recent years, deep convolutional neural networks (CNNs) have been developed well and achieved excellent performance improvements in various fields [2], including remote sensing scene classification. However, high-resolution remote sensing data still face challenges, including complex scenes and differences in target sizes (shown as Fig. 1) [3]. In detail, several confusing objects appear in the bridge scene, including building, trees, road, and river. In addition, a remote sensing image usually contains multiple objects whose sizes are quite different. For example, an airplane may occupy hundreds of pixels or only a few pixels in the remote scene. During the convolution process, the features of small objects may be lost, thus affecting the accuracy of the recognition results. Therefore, effectively dealing with or alleviating the problems above has become a key research direction of remote sensing scene classification.

In computer vision (CV), multiscale feature representations often highlight their importance in mitigating object size differences [4]. There are several classical methods, including image pyramid [5] and the spatial pyramid methods [such as spatial pyramid pooling networks (SPPNet) [6] and pyramid scene parsing network (PSPNet) [7]]. In addition, the widely used feature pyramid network (FPN) [8] makes the feature maps of different scales in each layer with strong semantic information. Importantly, it can fuse the features with low-resolution and high-resolution feature maps with less increasing computation.

The spatial multiscale representation methods can meet or alleviate the challenges of complex scenes and differences in target sizes [9]. However, the high-frequency details are lost due to convolution and downsampling operations in the spatial domain. Remote sensing images may lose the discriminative texture features. The enhanced multiscale texture feature in

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Fig. 1. Examples from the NWPU-RESISC45 dataset to illustrate the two main problems in remote sensing classification, including (a) interclass variance in size and (b) complex scene in remote sensing image. The scene objects are marked in the boxes.

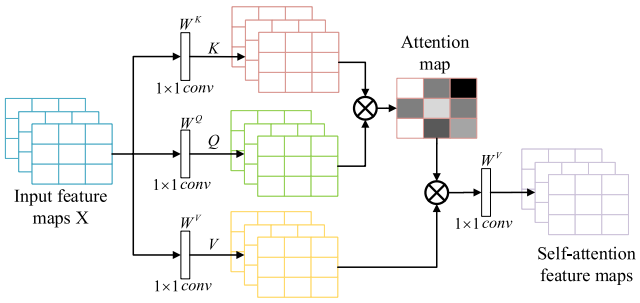


Fig. 2. Example of self-attention for the input feature maps. The 1 × 1 conv represents the convolutional operations with the kernel size 1 × 1.

the frequency domain is proposed to alleviate the problem above. As the simple and effective time–frequency analysis tool [10], [11], the wavelet transform is widely used for multiscale feature representation. It gradually refines the signal through the expansion and translation operations and finally achieves time subdivision of high frequencies and frequency subdivision of low frequencies. In addition, it can learn the explainable geometric texture features [12] in the frequency domain [13], which are very important in remote sensing scene recognition [12].

Regardless of the spatial- or frequency-domain multiscale representation methods, they have strong local representation capability. However, the global modeling capability is lacking but essential for the large scene. In recent years, the Transformer framework has been hotly researched, and it has made breakthroughs in natural language processing (NLP) and CV with its good global modeling capabilities. For example, it performs excellently on the ImageNet dataset and was once considered a deep learning framework that could surpass and replace CNNs. Besides, it has been widely used in remote sensing scene interpretation to face the complex remote sensing scenes [14].

In remote sensing scene classification, most work based on Transformer is mainly based on the typical visual Transformer ViT. It generates a feature map of a single low-resolution patch and then computes the long-range dependencies of each low-resolution region. However, it often restricts the model from learning the image's overall characteristics. Besides, it loses spatial structure features and ignores the high-resolution image features [15]. Self-attention is the core of the Transformer's global modeling, which can capture the long-term context relationship between pixels in the image and show certain performance advantages. Thus, some works are based on self-attention instead of the ViT framework, such as [15] and [16].

Besides, some researchers combine CNNs and Transformers to achieve significant performance improvement effectively, such as [17], [18], and [19]. They make full use of the efficiency of CNN in extracting local information and the ability of the Transformer to capture global information. In addition, some current works also consider the multiscale Transformer framework for remote sensing scene classification, such as [20]. However, they only consider the spatial-domain multiscale features. Studies suggest that the frequency domain can mine richer and discriminative image features.

Based on the research above, we construct an explainable multiscale Transformer framework in the spatial–frequency domain to obtain efficient remote sensing image feature representation. Some network layers and feature learning are explainable. In detail, its explainability is detailed as follows. First, the overall network framework is explained. The proposed spatial–frequency multiscale Transformer framework (SF-MSFormer) is designed to better learn the spatial and frequency multiscale features. It includes a spatial multiscale Transformer and a frequency multiscale Transformer to learn the spatial global multiscale features and the frequency global multiscale texture features. And then, the texture-enhanced encoder is explainable. It comprises the learnable lifting wavelet transform and self-attention. The network layer based on it is explainable to obtain and learn the shape and texture features. Besides, the inputs to the texture encoder are of the texture and shape prior, which can force self-attention to learn the global texture-enhanced features. Besides, the feature visualizations validate the explainability of the texture-enhanced encoder. Specifically, the frequency multiscale branch shows the texture-enhanced effects of the texture-enhanced encoder. This is explainable in feature maps.

Unlike previous work, the proposed framework considers global multiscale modeling in the spatial and frequency domains. Besides, the proposed texture-enhanced encoder can adaptively capture the global texture features, considering the differences in texture features at various resolutions. Finally, experiments are carried out on the benchmark remote sensing datasets to verify the effectiveness and convergence. In general, the main contributions of this article are as follows.

1) An explainable SF-MSFormer is proposed, which can adaptively capture the global spatial–frequency multiscale features.
2) The proposed spatial multiscale Transformer can learn the global multiscale features in the spatial domain, which can effectively aggregate the global context information between different scales.
3) The texture-enhanced encoder can learn the global enhanced texture feature and aggregate the frequency multiscale features. And then, the feature visualizations validate the explainability of it.
4) Experiments show that SF-MSFormer achieves competitive performance with good convergence on benchmark remote sensing datasets, including AID, UCM, WHU-RS19, and NWPU-RESISC45.

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The remainder of this article is organized as follows. Section II introduces the related works. Section III presents the proposed novel spatial–frequency multiscale Transformer for remote sensing classification. Section IV shows the experimental results and the analysis of the proposed method. Finally, Section V gives the concluding remarks of this article.

II. Related Works

In this article, the proposed spatial–frequency multiscale Transformer is designed for the remote sensing scene classification, mainly based on multiscale representation and the Transformer's global modeling ability. Therefore, this section mainly introduces the Transformer in remote sensing and the existing multiscale representation methods.

A. Transformer in Remote Sensing

The Transformer has achieved remarkable results in NLP [21], [22]. And then, it makes breakthroughs in CV with excellent global modeling capabilities. In addition, Transformer is extended to specific remote sensing tasks, including scene classification [16], [18], [23], semantic segmentation [24], [25], change detection [16], [23], and so on.

For remote sensing scene classification, most Transformer-related work is based on the ViT model. It divides the original image into multiple patches and encodes them as the ViT's input. However, it tends to limit the model to learning the image's overall features [15].

To deal with these limitations, there are some works combining CNNs for remote sensing classification, such as framework combined CNNs and vision transformer (CTNet) [19] and Resformer [17]. For example, CTNet [19] consists of a ViT stream (T stream) and a CNN stream (C stream). For T streams, flat image patches are sent to preprocessed ViT models to mine semantic features in high-resolution remote sensing images. The C stream uses the CNN to extract the structure feature as the supplementary of the T stream. A spatial-channel feature preserving vision transformer (SCViT) [26] is also proposed to retain the channel features in the ViT structure and consider the detailed geometric information images with high spatial resolution in the classification token.

In addition, some related work focuses on the core of the Transformer (self-attention, shown as Fig. 2). Chen et al. [27] learn the transform-invariant regions in combination with a Transformer-based attention mechanism to reduce the semantic gap effectively. Both [20] and [28] use the multiheaded attention mechanism. The former makes it obtain remote contextual relationships between pixels in an image. The latter uses it mainly to aggregate and explore the relevance of multilevel features.

Besides, some Transformer-based remote sensing methods consider scale representation. For example, Chen et al. [27] use the interscale loss between every two spatial transformer network flows. By introducing cross-scale loss, it can gradually learn the subtle differences in small regions of the image. In addition, an efficient multiscale transformer (EMST) is proposed, which can mine the abundant knowledge with various scales hidden in remote sensing scenes [20]. This provides evidence to construct the networks to exploit the rich features of remote sensing scenes fully.

B. Multiscale Representation

The existing multiscale representation methods are mainly divided into multiscale network-based and multiscale geometric analysis-based methods.

1) Multiscale Representation Based on Multiscale Networks: With the continuous development of neural networks, the depth and performance of networks such as AlexNet [29], VGG Nets [30], GoogLeNets [30], and ResNet [31] continue to increase. Multiscale features can improve the performance of neural networks. As for multiscale representation, convolution and pooling operations with different kernel sizes are the keys to obtaining multiscale features. There are several typical multiscale representation networks, including spatial pyramid pooling (SPP) [6], atrous spatial pyramid pooling (ASPP) [32], and image pyramid (Gaussian and Laplacian image pyramid [33].)

Among them, the image pyramid enhances multiscale changes but brings much calculation. Spatial pyramid methods (such as SPPNet [6] and PSPNet [7]) use features of different scales for prediction. However, there is insufficient semantic information in the low-level feature maps and high-level feature maps with low resolution. In contrast, the FPN can handle the multiscale change problem while adding minimal computation. It enables feature maps of different scales in each layer to have strong semantic information and simultaneously fuses feature maps with strong low-resolution semantic information and high-resolution feature maps with weak semantic information but rich spatial information. Besides, it became a classic example of dealing with different target sizes, such as [34], [35], and [36], especially for small target recognition [37]. Importantly, it can fuse the features with low-resolution and high-resolution feature maps with less increasing computation.

2) Multiscale Representation Based on Multiresolution Analysis: The multiscale geometric analysis methods are practical tools for multiresolution analysis. The existing multiscale geometric analysis methods mainly include wavelet transform, Ridgelet transform, Contourlet transform [38], and so on. They can effectively capture different image geometric features. Because the above transformations all have basis functions of different scales, image features of different scales can often be obtained. Traditional multiscale geometry is often used to obtain multiscale and multidirectional features of images and is applied to tasks such as recognition [39], detection [40], and denoising [41]. However, the performance of pure multiscale methods is subpar on deep learning tasks.

With the continuous development of deep learning, multiscale geometric methods are also effectively combined with neural networks for deep learning tasks, such as image classification WaveletCNN [42], contourlet convolutional neural networks (C-CNN) [43], image denoising WaveCNets [44], and image super-resolution deep shearlet residual learning network (DSLRN) [45]. The related methods mainly integrate the multiscale geometric analysis and CNN features, such as Scattering Network [46], WaveletCNN [42], and C-CNN [43]. The above multiscale geometric methods combined with neural networks have achieved relatively excellent results in deep learning tasks, providing a direction for the subsequent combination of multiscale geometry and neural networks.

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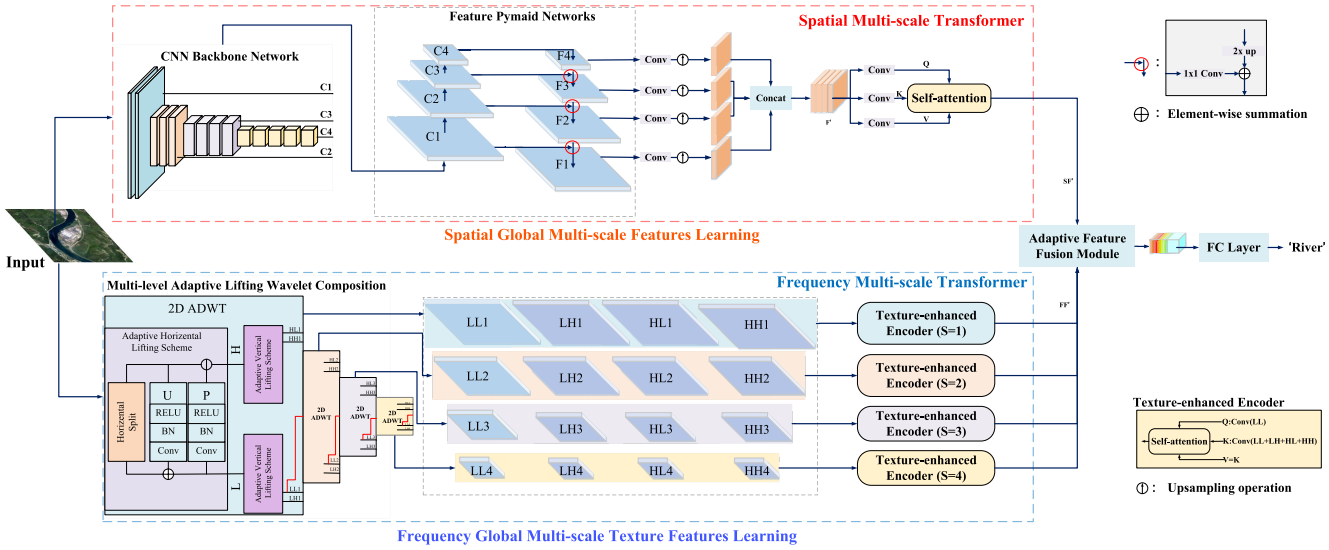


Fig. 3. Overall framework of the proposed multiscale spatial–frequency Transformer for remote sensing scene classification. It consists of two branches: the spatial and frequency multiscale Transformer branches. $S = i, i = 1, 2, 3, 4$ represents the four detailed scales.

III. Explainable Spatial–Frequency Multiscale Transformer

The explainable spatial–frequency multiscale Transformer, namely, SF-MSFormer, is proposed, which is a joint spatial and frequency multiscale representation (FMSR) framework. Besides, effective global modeling and an adaptive combination of spatial–frequency multiscale features improve its performance in remote sensing scene classification. The proposed SF-MSFormer is detailed in this section.

A. Overall Framework

The overall framework of the proposed spatial–frequency multiscale Transformer is shown in Fig. 3. It depicts the multiscale representation in the spatial and frequency domains with the global modeling ability. It mainly consists of a spatial multiscale Transformer branch and a frequency multiscale Transformer branch.

In the spatial multiscale Transformer branch, the FPNs help obtain the spatial multiscale features. And then, the multiscale spatial features are aggregated after convolution and upsampling operations. The aggregated features are sent to the self-attention module to obtain the multiscale features' interglobal dependence.

In the frequency branch, the FMSR is obtained by the multilevel adaptive lifting wavelet composition. And then, different scale features are sent to the texture-enhanced encoder, which can capture the texture feature at different scales. In addition, an adaptive feature aggregation module is designed to obtain the collaborative features of the spatial and the frequency multiscale Transformer branch. Finally, the collaborative features are sent to the classifier for final recognition. In detail, the overall learning procedure of SF-MSFormer is listed in Algorithm. 1.

Algorithm 1 Learning Procedure of SF-MSFormer

Input: Training dataset: $\mathbf{x} = \{x_n | n = 1, 2, \ldots, N\}$, the corresponding labels: $\mathbf{y} = \{y_n | n = 1, 2, \ldots, N\}$; the number of categories $\mathbf{T}$.

Output: Classification result $\hat{\mathbf{y}} = \{\hat{y}_n | n = 1, 2, \ldots, N\}$.

Preparation: Calculate the adaptive wavelet transform composition times M by (17).

Begin

for n *in* $[1, N]$ **do**

- Given an input image x_n, obtain the aggregated spatial multiscale pyramid feature F';
- Calculate $x_n \rightarrow F'$ by (1) and $x_n \rightarrow LL1, LH1, HL1, HH1$ by (4–8);
- **for** i *in* $(1, M]$ **do**
  - Calculate the frequency feature of different scales LLi, LHi, HLi, HHi by (9);
  - Calculate Qi, Ki, Vi by (11), and then calculate the texture-enhanced feature at scale i, $FFi : softmax((QiKi^T)/\sqrt{d})Vi$ and obtain the aggregated FF' by (14).
- Calculate $F' \rightarrow Q', K', V'$ by (12), and then calculate $SF' : softmax((Q'K'^T)/\sqrt{d})V'$;
- Adaptive feature fusion operations for two branches' output as $f1 : C_1 \otimes SF'$, $f2 : C_2 \otimes FF'$;
- Cascade the feature maps OF: $f1 \circ f2$ for final recognition;
- Minimize (19) and update the parameters by the SGD optimizer until convergence.

B. Spatial Multiscale Transformer

The spatial multiscale Transformer branch is aimed at learning the global spatial multiscale features. The CNNs' backbone is used to obtain the pyramid network feature $C1, C2, C3, C4$. The multiscale feature at different levels is obtained, including $F1, F2, F3, F4$. The aggregated features F' are formulated as

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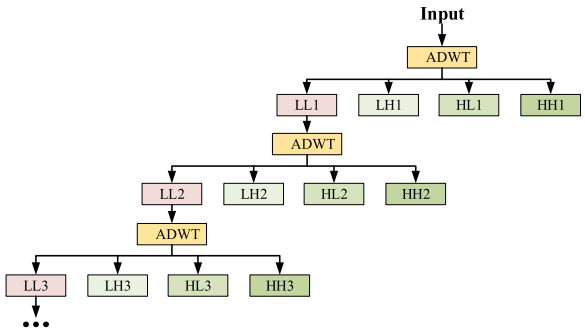


Fig. 4. Process of multilevel adaptive lifting wavelet transform.

follows:

$$F' = F1 \circ \mathrm{Conv}(F2) \uparrow^1 \circ \mathrm{Conv}(F3) \uparrow^2 \circ \mathrm{Conv}(F4) \uparrow^3 \quad (1)$$

where Conv represents the convolutional operation with kernel $= 3 \times 3$, $\uparrow$ represents the upsampling operations, and $\circ$ represents the concatenate operation. In detail, $\uparrow^1$ means 1×2 times upsampling. $\uparrow^2$ means 2×2 times upsampling. $\uparrow^3$ means 3×2 times upsampling. And then, the aggregated multiscale feature F' is the self-attention module's input. Q', K', and V' can be formulated as

$$Q' = W^q F' K' = W^k F' V' = W^v F'. \quad (2)$$

Thus, the spatial multiscale self-attention feature maps can be formulated as

$$\mathrm{SF}' = \mathrm{SAtt}\left(Q', K', V'\right) = \mathrm{softmax}\left(\left(Q' K'^T\right)/\sqrt{d}\right) V' \quad (3)$$

where $\sqrt{d}$ we adopted is a learnable parameter, and SAtt represents the traditional self-attention operation.

C. Frequency Multiscale Transformer

The frequency multiscale Transformer branch is aimed at learning the global frequency multiscale texture features. Its framework is shown in Fig. 3. It mainly consists of FMSR and the texture-enhanced encoder. The FMSR is mainly obtained by the multilevel adaptive lifting wavelet transform, shown as Fig. 4.

1) Adaptive Discrete Wavelet Transform: The adaptive discrete wavelet transform (ADWT) is mainly based on the lifting wavelet transform [47]. It inherits the above advantages of the wavelet transform. In addition, it allows for adaptive wavelet transforms so that we can obtain the multiresolution feature from the coarsest levels to the more refined levels by refining only in the areas of interest [48], [49]. The general framework of the lifting wavelet transform consists of three stages: Split, Predict, and Update. A whole family of biorthogonal wavelets can be constructed by varying the three stages of lifting. The predictor and updater operations make it flexible to design the lifting wavelets.

For an input X, the low-pass component L and the high-pass component H can be formulated as

$$\begin{aligned} \left\{L^0, H^0\right\} &= S_h\{X\} \\ H &= H^0 - P\left(L^0\right) \\ L &= L^0 + U\left(H^0\right) \end{aligned} \quad (4)$$

where S_h represents the initial horizontal split, and H^0 and L^0 represent the initial splitting result. P and U represent the update and prediction operations, respectively. Thus, the adaptive wavelet transform can obtain the low-pass component LL and the low-pass components HL, LH, and HH as

$$\begin{aligned} \left\{LL^0, LH^0\right\} &= S_v\{L\} \\ LH &= LH^0 - P\left(LL^0\right) \\ LL &= LL^0 + U\left(LH^0\right) \end{aligned} \quad (5)$$

$$\begin{aligned} \left\{HL^0, HH^0\right\} &= S_v\{H\} \\ HH &= HH^0 - P\left(HL^0\right) \\ HL &= HL^0 + U\left(HH^0\right) \end{aligned} \quad (6)$$

where S_v represents the vertical splitting operations.

Developing predictor and updater with simple wavelet functions only obtains some special wavelets, which lacks strong universality for diverse data. Thus, adopting the convolution operation as the predictor and updater can make it adaptive to capture the low-pass and high-pass features [50]. Therefore, the convolutional operation with batch normalization (BN) and rectified linear unit (RELU) is adapted as the updater and the predictor to obtain the adaptive frequency multiscale features. For signal x, there is

$$P = U = \delta(f(\delta(f(x)))) \quad (7)$$

where δ represents the RELU function, and $f()$ represents the convolution operation.

Based on the adaptive wavelet transform, the multiscale frequency features can be formulated as follows. As shown in Fig. 4, for the input image X, the frequency features at scale $= 1$ can be formulated as

$$\begin{aligned} \mathrm{Scale1}_{\mathrm{Fre}_1,\ldots,\mathrm{Fre}_4} &= \mathrm{ADWT}(X) \\ &= [LL1, LH1, HL1, HH1]. \end{aligned} \quad (8)$$

The frequency features of scale $= \{i | i = 2, 3, .., M\}$ can be formulated as

$$\begin{aligned} \mathrm{Scale}i_{\mathrm{Fre}_1,\ldots,\mathrm{Fre}_4} &= \mathrm{ADWT}(LLi) \\ &= [LLi, LHi, HLi, HHi]. \end{aligned} \quad (9)$$

2) Texture-Enhanced Encoder: As shown in Fig. 5, the frequency self-attention takes the multifrequency features into consideration, different from the spatial self-attention. The ADWT's output can capture the low-pass and high-pass frequency features. The frequency multiscale features can be obtained as Fig. 4. Besides, the high-pass can better curve the texture information of the input. Thus, effectively using the frequency feature and self-attention can achieve the texture-enhanced features.

First, the multiscale frequency features can be extracted by the multilevel adaptive lifting wavelet transform. And then, the proposed texture-enhanced encoder is used for the texture-enhanced features. The differences in self-attention and the texture-enhanced encoder are shown as Fig. 6.

For the input X, the proposed texture-enhanced encoder at scale $= 1$ can be formulated as

$$\mathrm{FF1} = \mathrm{SAttT}(\mathrm{ADWT}(X)) \quad (10)$$

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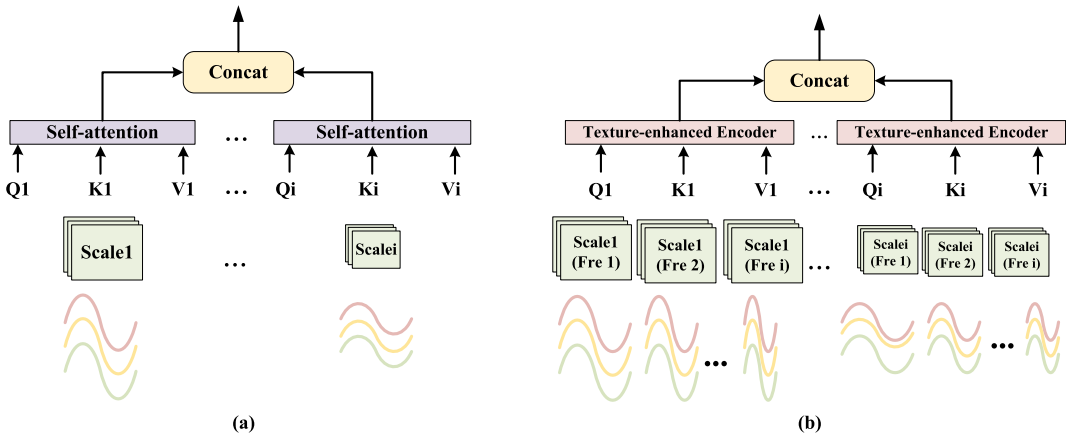


Fig. 5. Comparison of spatial and frequency multiscale self-attention. (a) Spatial multiscale self-attention. (b) Frequency multiscale self-attention. Different waves represent the features at different scales and frequencies.

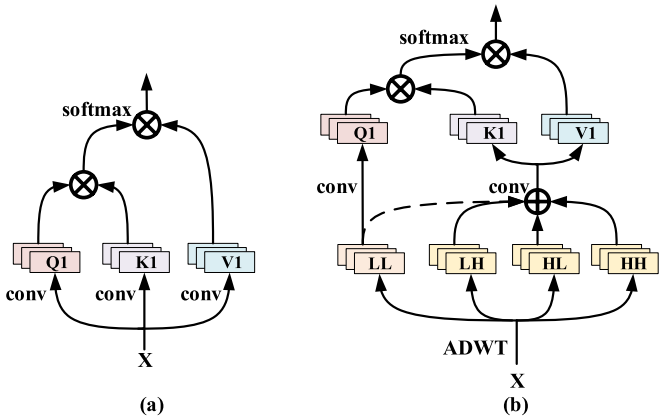


Fig. 6. Comparison of the proposed texture-enhanced encoder with the traditional self-attention. (a) Self-attention. (b) Texture-enhanced encoder.

where ADWT can be calculated by (8). SAttT represents the texture-enhanced self-attention operation. In detail, the query $(Q1)$, key $(K1)$, and value $(V1)$ are formulated as follows:

$$\begin{aligned} Q1 &= \text{Conv}(LL1) \\ K1 &= \text{Conv}(LL1 + LH1 + HL1 + HH1) \\ V1 &= K1. \end{aligned} \tag{11}$$

In contrast, the frequency self-attention takes the learnable frequency features as the input of self-attention, which adopts the geometry structure as the prior knowledge. Equation (10) can be formulated as

$$\begin{aligned} \text{FF1} &= \text{SAtt}(Q1, K1, V1) \\ &= \text{softmax}\left(\left(Q1K1^T\right)/\sqrt{d_1}\right)V1 \end{aligned} \tag{12}$$

where $\sqrt{d_1}$ is a learnable parameter.

Different from the spatial multiscale Transformer, the frequency multiscale adapts the self-attention for different scale features. The scale $= 2, 3, 4$ frequency multiscale features are formulated as

$$\begin{aligned} \text{FF2} &= \text{SAtttT}(\text{ADWT}(LL1)) \\ \text{FF3} &= \text{SAttT}(\text{ADWT}(LL2)) \\ \text{FF4} &= \text{SAttT}(\text{ADWT}(LL3)). \end{aligned} \tag{13}$$

And then, the output FF$'$ of the frequency multiscale Transformer considers FF1, FF2, FF3, and FF4 as the FMSR features

$$\text{FF}' = C1 \otimes \text{FF1} \circ C2 \otimes \text{FF2} \circ C3 \otimes \text{FF3} \circ C4 \otimes \text{FF4} \tag{14}$$

where $C1, C2, C3, C4$ represents the convolutional kernel calculated by the adaptive feature fusion strategy.

D. Adaptive Feature Fusion Module

Because the output of the spatial and frequency multiscale Transformer branches is in different size. Thus, an adaptive feature fusion module is presented for feature aggregation. The spatial multiscale feature $[\text{SF}'(w1 \times h1 \times c1)]$ and the frequency multiscale feature $[\text{FF}'(w2 \times h2 \times c2)]$ can be aggregated to the output feature OF. It is formulated as

$$\text{OF} = C_1 \otimes \text{SF}' \circ C_2 \otimes \text{FF}' \tag{15}$$

where C_1 and C_2 are the adaptive convolution operations with stride. It can make the features into the same size for adaptive feature concatenation. For example, if keeping the output size as weight (w) $\times$ height (h), the adaptive convolution ac_1 and ac_2 can be formulated as

$$\begin{aligned} \text{kernel_size}_i &= wi - (w-1) * \text{stride} \\ \text{stride}_i &= \text{floor}((wi/w)) \end{aligned} \tag{16}$$

where $i = 1, 2$. Finally, the output OF is in the size of $w \times h \times (c1 + c2)$ after the adaptive feature fusion module. $(c1 + c2)$ represents the output channel summation of the convolution ac_1 and ac_2.

E. Network Details

1) CNNs' Backbone: The pretrained ResNet18 and ResNet34 are mainly adopted as the CNNs' backbone in the spatial multiscale Transformer to generate the spatial multiscale features. Besides, upsampling is adopted in the feature pyramid. The nearest neighbor interpolation method is selected as the concrete implementation of upsampling.

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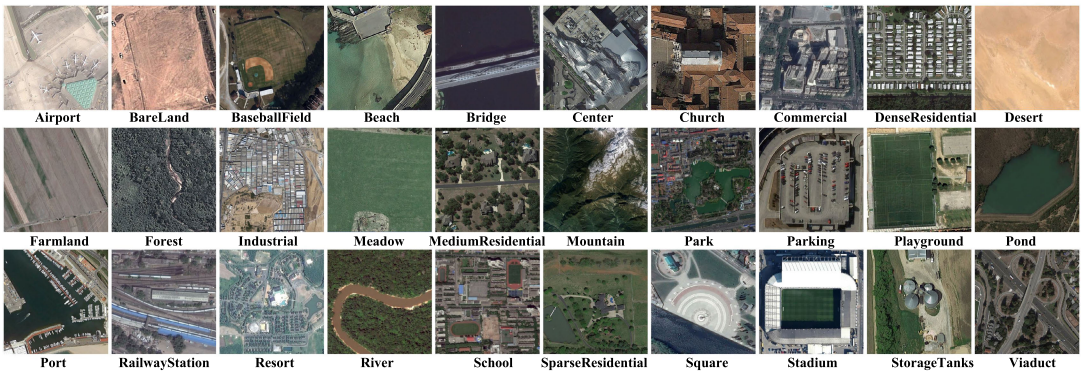


Fig. 7. Example samples of the AID dataset.

2) Number of the Scale in Frequency Multiscale Transformer Branch: The adaptive wavelet transform is mainly used to obtain the multiscale frequency features. As for the adaptive wavelet transform, it can be decomposed at M times, which is formulated as

$$M = \log_2 N - 2 \tag{17}$$

where $N \times N$ represents the input's size. Thus, if the input is of the size 224×224, it can be decomposed into five scales. Here, four scales are mainly concerned with our overall framework, as shown in Fig. 3.

3) Loss Function: The cross-entropy loss is chosen to optimize the network parameters in the training stage. For the sample x and the label y, the softmax classifier can calculate the probability that the input x is predicted to each true label as

$$p(y|x) = \frac{e^{f_y}}{\sum_{c=1}^{C} e^{f_c}} \tag{18}$$

where f_i represents the prediction value. And the loss function can be formulated as

$$\text{loss} = -q \sum \log p(y|x) + (1-q) \log \sum (1 - p(y|x)) \tag{19}$$

where q is the true label value with one-hot coding, and $q = [x_1, x_2, \ldots, x_C]$, $x_i = 1(i \in [0, C])$, others $= 0$.

IV. Experiments

In this section, the experimental settings are introduced first. And then, the experimental results verify the effectiveness of the proposed SF-MSFormer. In addition, the ablations explore the importance of the components of SF-MSFormer. Finally, the visualization and analysis are shown to perform the effects of texture enhancement.

A. Experimental Settings

1) Datasets: Four datasets for remote sensing scene classification are mainly adopted in our experiments, including the AID [51], UCM [52], WHU-RS19 [53], and NWPU-RESISC45 datasets [54]. The characteristics of these four datasets are shown in Table I.

1) *AID [51]:* It is an aerial imagery dataset that collects samples from Google Earth images. There are 10 000 images with 30 categories of remote sensing scenes. There are 50 images in each category. All the pictures are in the size of 600×600 and with a spatial resolution of up to 0.5 m. The example samples of AID are shown in Fig. 7.
2) *UCM [52]:* It is obtained from the Google satellite imagery, which consists of 2100 images with 21 categories of remote sensing scenes. There are 100 images in each category. They are all at the size of 256×256 and with a spatial resolution at 0.3 m. The example samples of UCM are shown as Fig. 8.
3) *WHU-RS19 [53]:* It is obtained from the USGU National Map Urban Area Image Collection, which includes 1005 images with 19 categories of remote sensing scenes. There are about 200–400 images in each category. They are all at the size of 600×600 and with a spatial resolution of up to 0.5 m. The example samples of WHU-RS19 are shown in Fig. 9.
4) *NWPU-RESISC45 [54]:* It was released by Northwestern Polytechnical University in 2016 and extracted from remote sensing images of land use types of Google Earth. Specifically, there are a total of 31 500 images. It contains 45 categories, and each class corresponds to 700 remote sensing images. The size of each picture is 256×256, and the resolution range is 0.2–30 m. Some examples are shown as Fig. 1.

TABLE I
Characteristics of the AID, UCM, and WHU-RS19 Datasets

Datasets	Images per class	Scene class	Total images	Spatial resolution(m)	Image sizes
AID	200∼400	30	10000	0.5m∼0.8m	600×600
UCM	100	21	2100	0.3 m	256×256
WHU-RS19	∼50	19	1005	up to 0.5m	600×600
NWPU-RESISC45	700	45	31,500	0.2m∼30m	256×256

In addition, all the datasets are augmented with random cropping and horizontal flipping operations for training. Besides, data normalization is used for all the datasets. The experimental datasets are all resized to 224×224 before being fed into the network.

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Fig. 8. Example samples of the UCM dataset.



Fig. 9. Example samples of the WHU-RS19 dataset.

2) Settings: The **bilinear** upsampling method is used in the spatial multiscale Transformer. The adaptive average pooling with output size 1×1 is adopted for the adaptive feature fusion module. In addition, the stochastic gradient descent (SGD) optimizer is used to train the SF-MSFormer for remote sensing scene classification with a momentum of 0.9.

All the experiments have been trained for 300 epochs with the decay of the learning rate at epochs 150 and 200. The initial learning rate is set at 0.01, and the batch size is set at 32 for training. All the experiments have been done on the NVIDIA Titan V100 four GPUs with PyTorch 1.7.0.

3) Evaluation Metrics: The proposed method mainly experiments with the remote sensing scene classification datasets. Thus, the commonly used two evaluation metrics are adopted to determine the effectiveness of the proposed methods in our experimental results, including the overall accuracy (OA) and the confusion matrix. It should be noted that all the experiments have been conducted three times, and the average value (Ave.) and the standard deviation (std.) of the three-time experimental results were counted during the experiment. In addition, the parameters and GFlops are selected as indicators to evaluate the models' computation complexity.

4) Contrasting Methods: The proposed SF-MSFormer is a spatial-FMSR network with self-attention. It integrates CNNs and multiresolution analysis features with the self-attention module. Thus, there are several type of related methods chosen as the compared methods, including multiscale-related, CNNs-based, Transformer-based methods, and frequency CNNs.

The CaffeNet [55], GoogleNet [55], and VGG-VD-16 [54] are three classical CNN-based methods. CNNs-AE [56] and CNNs-NN [56], respectively, represent the method that combines convolutional features with the sparse autoencoder and the method that extracts the feature representation by the pretrained CNNs following one hidden layer neural network. The global feature + MLP (GM) [57] method represents the method that combines the global feature with MLP. GLM16 [57] and GLM512 [57], respectively, represent the GM method with 16 selected local features or 512 local features. Gradient boosting random convolutional network (GBRCN) [58] combines different deep neural networks via a boosting way. Discriminant correlation analysis for feature fusion (DCA-Fusion) [59], compressive sensing multifeature (CS-Fusion) [60], and lightweight CNN classification method based on branch feature fusion (LCNN-BFF) [61] are fusion networks. DCA-Fusion [59] fuses different layers' features from the original features extracted from VGG-Net. CS-Fusion [60] fuses the multifeature for the land-use classification task. LCNN-BFF [61] proposes the branch feature fusion convolution network.

Besides, SF-MSFormer uses the FPN in the network design. Thus, the performance of the FPN-related method, EFPN-DSE-deep feature fusion (TDFF) [62], is listed here. It proposes to use the enhanced FPN (EFPN) with deep semantic embedding (DSE) for remote sensing scene classification. In addition, SF-MSFormer is a multiscale and frequency-representation-related method. Thus, some multiscale-related and frequency convolutional networks [such as multi-scale staking attention pooling (MS2AP) [63], multiscale feature fusion covariance network (MF^2CNet) [64], robust space–frequency joint representation (RSFJR) [65], Wavelet CNN [43], and C-CNN [43]] are selected as the contrasting methods. MS2AP [63] proposes multiscale stacking attention pooling. MF^2CNet [64]) considers multiscale feature fusion covariance network with octave convolution. Wavelet CNN and C-CNN [43] are two multiresolution methods that consider multiscale frequency features. RSFJR [65] is a space–frequency joint representation method for remote sensing image scene classification. In addition, three Transformer-based methods including ViT [55], SCViT [55], and efficient multiscale transformer and cross-level attention learning (EMTCAL) [20] are shown as the compared methods too.

B. Experimental Results

1) Results on AID: As shown in Table II, the experimental results of the related algorithms are presented when 50% of the AID dataset is used for training. $\text{ResNet18}_{\text{global+local}}$ performs 96.76% OA, which achieves better performance in the compared CNN methods. In addition, the Transformer-based methods, including ViT, EMTCAL, and SCViT, show about 96.49%, 97.49%, and 96.98% average precision. The latest novel multiscale Transformer with cross-attention method, EMTCAL, shows the best performance with 0.73% increased accuracy than $\text{ResNet18}_{\text{global+local}}$. SF-MSFormer with the ResNet34 backbone offers about 1.23% increased precision than EMTCAL on the AID dataset, respectively. In addition, SF-MSFormer with the ResNet34 backbone shows about 0.25% OA than that with the ResNet18 backbone.

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TABLE II

CLASSIFICATION ACCURACY (%) ON THE AID, UCM, AND WHU-RS19 DATASETS WITH 50% FOR TRAINING, INCLUDING THE PUBLICATION. THE BEST RESULTS HAVE BEEN HIGHLIGHTED IN BOLD. * REPRESENTS THE FINE-TUNED MODEL

Datasets / Methods	Publication (Years)	AID Ave.	AID Std.	WHU-RS19 Ave.	WHU-RS19 Std.	UCM Ave.	UCM Std.	Parameters Unit(M)	GFlops Unit(G)
GBRCN [58]	TGRS (2015)	91.40	1.69	91.34	0.98	91.78	2.34	-	-
CS-Fusion [60]	GRSL (2015)	90.06	2.07	89.27	0.91	90.65	1.48	-	-
CNN-AE [56]	IJRS (2016)	92.03	1.36	92.58	1.07	92.06	1.57	-	-
CNN-NN [56]	IJRS (2016)	92.97	1.45	93.31	1.24	93.49	2.54	-	-
CaffeNet [51]	TGRS (2017)	89.53	0.31	96.24	0.67	93.98	0.67	-	-
GoogleNet [51]	TGRS (2017)	86.39	0.55	94.71	0.75	92.70	0.6	7	0.75
VGG-VD-16 [51]	TGRS (2017)	89.64	0.36	96.05	0.42	94.14	0.69	138.36	7.75
DCA-Fusion [59]	TGRS (2017)	94.33	1.52	93.56	0.96	93.84	0.79	-	-
GM [57]	TGRS (2018)	89.22	1.86	88.15	0.88	88.77	1.55	-	-
GLM512 [57]	TGRS (2018)	91.98	1.34	90.23	0.79	90.12	1.27	-	-
GLM16 [57]	TGRS (2018)	93.45	1.52	92.99	0.83	92.60	1.31	-	-
Yuan et.al [57]	TGRS (2018)	95.29	1.28	95.76	0.74	94.97	1.16	0.6	-
APDC-Net [66]	IGRSL (2019)	92.15	0.29	-	-	95.01	0.43	-	-
RSFJR[65]	TGRS(2019)	96.18	1.36	97.48	0.85	97.21	0.65	550.98	-
Wavelet CNN[43]	TNNLS(2020)	96.65	0.24	86.2	0.68	92.19	0.47	10.24	2.98
C-CNN[43]	TNNLS(2020)	97.36	0.45	-		-	-	-	-
LCNN-BFF [61]	J-STARS (2020)	94.62	0.16	-	-	94.64	0.21	6	-
EFPN-DSE-TDFF [62]	TGRS (2021)	94.50	0.3	-	-	96.19	0.13	-	-
MS2AP[63]	Neurocomputing(2021)	96.46	0.20	**98.88**	0.40	-	-	23.7	-
ResNet18 [3]	TNNLS (2022)	95.51	0.31	-	-	97.43	0.19	11.19	1.84
ResNet18$_{global+local}$ [3]	TNNLS (2022)	96.76	0.20	-	-	97.95	0.05	-	-
ViT [20]	TGRS (2022)	96.49	0.18	-	-	98.48	0.39	49	10.1
SCViT [26]	TGRS (2022)	96.98	0.16	-	-	98.9	0.19	40.1	9.3
MF^{2}CNet[64]	TGRS(2022)	97.02	0.28	-	-	98.76	0.18	30.12	2.42
EMTCAL* [20]	TGRS (2022)	97.49	0.36	97.6	0.23	98.67	0.16	27.0	4.23
SF-MSFormer (ResNet18)	(2022)	**98.57**	0.23	97.8	0.39	**99.35**	0.26	26.12	19.56
SF-MSFormer (ResNet34)	(2022)	**98.72**	0.31	**98.6**	0.23	**99.72**	0.12	36.31	21.41

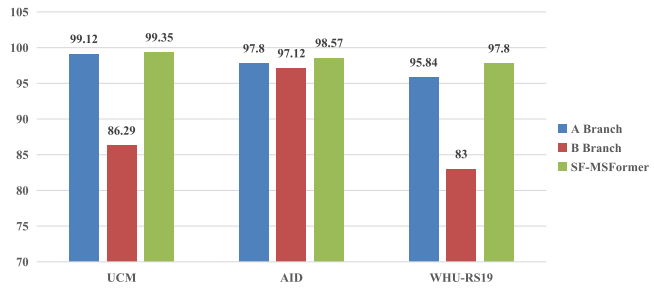


Fig. 10. Compare precision (%) of the spatial multiscale representation branch (A) and the FMSR branch (B).

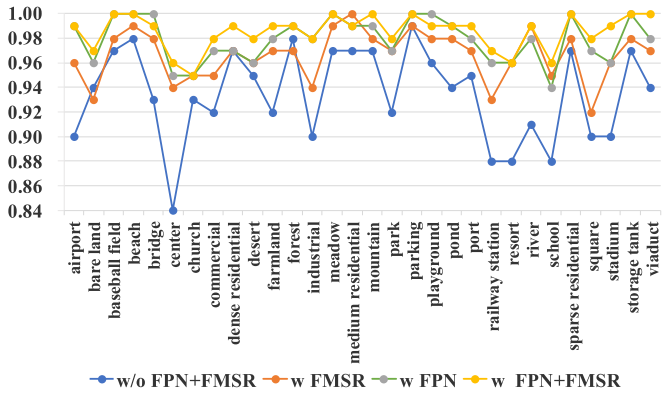


Fig. 11. Compare precision curves (%) about the SF-MSFormer with (w) or without (w/o) spatial (FPN) and frequency multiscale features (FMSR).

2) Results on UCM: As shown in Table II, the experimental results of the related algorithms are presented when 50% of the UCM dataset is used for training. Among the compared methods, ResNet18$_{\text{global+local}}$, EMTCAL* [20], ViT [20], MF^{2}CNet [64], and SCViT [26] published in 2022 shows great improvements in performance than the methods before. Among them, SCViT [26] won the first place in the contrasting methods with 98.90% average OA on the UCM dataset. In contrast, the proposed SF-MSFormer arrives at the state-of-the-art with 99.72% average OA. It improves about 0.82% increased performance than SCViT. Besides, SF-MSFormer with the ResNet34 backbone still shows better performance than that of the ResNet18 backbone.

3) Results on WHU-RS19: As shown in Table II, the experimental results of the related algorithms are presented when 50% of the WHU-RS19 dataset is used for training. Among the compared methods, CaffeNet [51], Yuan et al. [57] and VGG-VD-16 [51] show better performance in the CNNs. Compared with the related works, SF-MSFormer shows about increment precision from 1.56% to 9.65%. The Transformer-based method, EMTCAL, shows the best accuracy in the contrasting methods. It achieves about 1.41% increased OA by the CaffeNet [51]. As shown, it suggests that the proposed SF-MSFormer with ResNet34 achieves 1.0% increased precision than the contrasting methods mentioned above. Meanwhile, SF-MSFormer with ResNet34 performs better than that with the ResNet34 backbone.

In addition, 20%, 50%, and 80% of the datasets are random split for training the proposed model, as shown in Table III. When 20% of the dataset is used for training, the proposed SF-MSFormer shows 95.25%, 96.68%, and 96.63% average accuracy on the UCM, AID, and WHU-RS19 datasets, respectively. When 50% of the dataset is used for training, the

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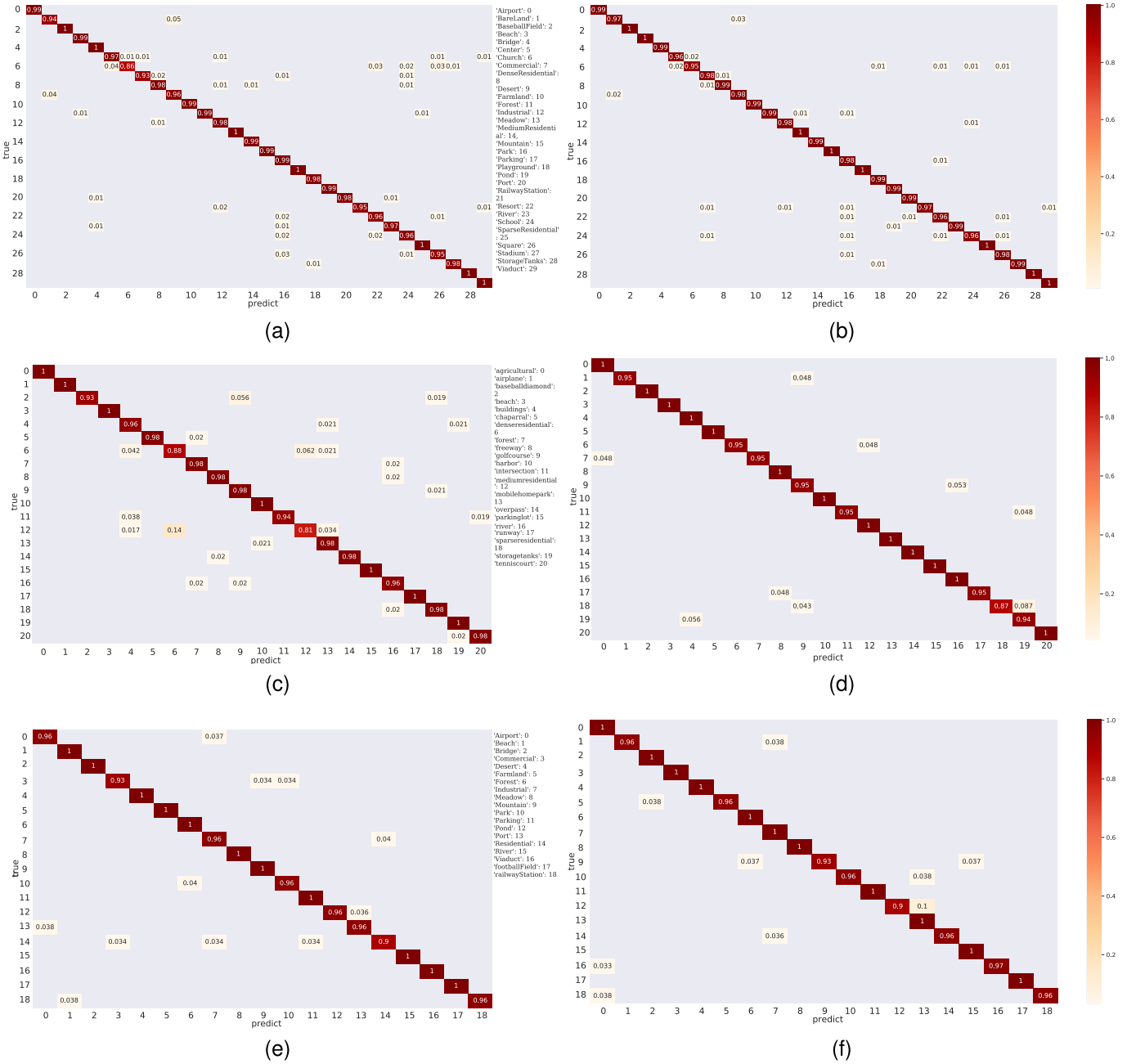


Fig. 12. Confusion matrices of the proposed methods on the AID, UCM, and WHU-RS19 datasets (50% for testing). (a) SF-MSFormer without self-attention on AID. (b) SF-MSFormer on AID dataset. (c) SF-MSFormer without self-attention on UCM. (d) SF-MSFormer on UCM. (e) SF-MSFormer without self-attention on WHU-RS19. (f) SF-MSFormer on WHU-RS19.

TABLE III

DIFFERENT RATIOS OF THE DATASETS FOR TRAINING (%). THE BEST RESULTS HAVE BEEN HIGHLIGHTED IN BOLD

Datasets	UCM		AID		WHU-RS19	
Ratio	Ave.	Std.	Ave.	Std.	Ave.	Std.
20%	95.25	0.45	96.68	0.64	96.63	0.23
50%	99.35	0.35	**98.57**	0.23	97.80	0.39
80%	**99.82**	0.46	98.12	0.62	**98.54**	0.53

proposed SF-MSFormer shows 99.35%, 98.57%, and 97.8% average accuracy on the UCM, AID, and WHU-RS19 datasets, respectively. When 80% of the dataset is used for training, the proposed SF-MSFormer shows 99.82%, 98.12%, and 98.54% average accuracy on the UCM, AID, and WHU-RS19 datasets, respectively. When the ratio of the training data is increased, it shows increased average precision on the UCM and WHU-RS19 datasets except for the AID dataset.

4) Results on NWPU-RESISC45: Table IV shows the performance of the proposed SF-MSFormer and some state-of the-art methods. About 10% and 20% of the dataset are mainly considered as the training set. The traditional CNNs, including AlexNet, VGGNet-16, ResNet18, and ResNet 50, are fine-tuned to gain the performance on the NWPU-RESISC45 dataset. Among them, ResNet18 shows better performance. Besides, ResNet18$_{\text{global+local}}$ [3] shows about 1.13% and 1.18%

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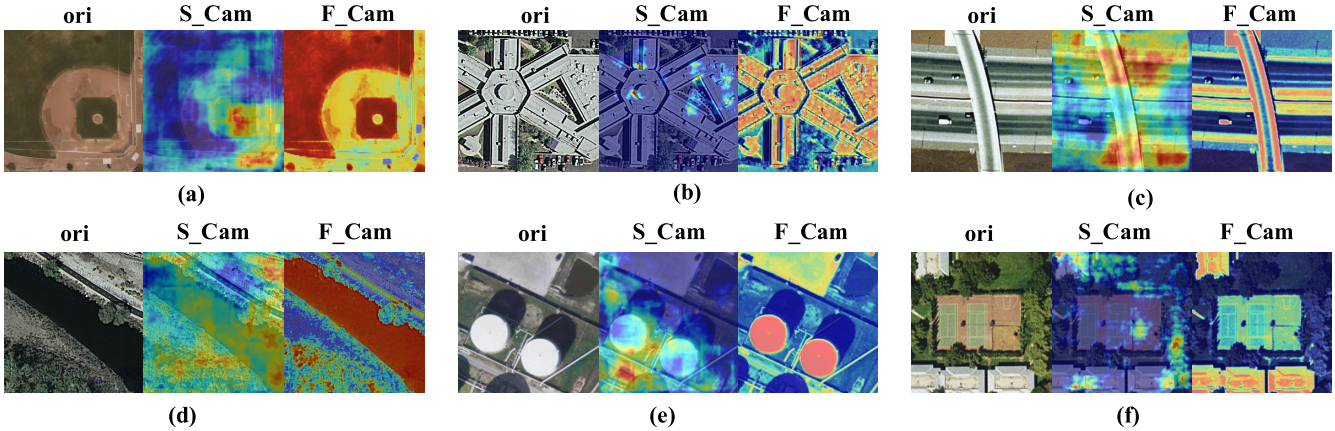


Fig. 13. Category activation map (Cam) feature visualizations of the proposed SF-MSFomer about six examples, including (a) baseballdiamond, (b) buildings, (c) overpass, (d) river, (e) storagetanks, and (f) tenniscourt. ori represents the original image. S_Cam and F_Cam represent the spatial multiscale Transformer and the frequency multiscale Transformer branches' feature category activation map, respectively.

TABLE IV

CLASSIFICATION RESULTS (ACCURACY %) ON THE NWPU-RESISC45 DATASET. THE BEST RESULTS HAVE BEEN HIGHLIGHTED IN BOLD

Methods	10% for training	20% for training
Fine-tune AlexNet [51]	81.22±0.19	85.16±0.18
Fine-tune VGGNet-16 [51]	87.15±0.45	90.36±0.18
Fine-tune ResNet50 [51]	88.48±0.21	91.86±0.19
ResNet18 [3]	88.91±0.23	91.77±0.18
ResNet18$_{global+local}$[3]	90.04±0.15	92.95±0.09
LCNN-BFF [61]	86.53±0.15	87.84±0.26
Attention GANs [67]	86.11±0.22	89.44±0.18
Two-stream fusion [68]	85.02±0.25	87.01±0.19
ADFF [69]	90.58±0.19	91.91±0.23
WaveletCNN [43]	84.69±0.44	88.84±0.38
C-CNN [43]	85.93±0.51	89.57±0.45
MS2AP[63]	91.86±0.18	93.44±0.18
MF^{2}CNet [64]	92.07±0.22	93.85±0.27
ViT [20]	90.05±0.29	92.61±0.14
EMTCAL [20]	91.63±0.19	93.65±0.12
SCViT [26]	92.72±0.04	94.66±0.1
SF-MSFormer(ResNet18)	91.84±0.35	94.37±0.32
SF-MSFormer(ResNet34)	**92.74±0.23**	**94.83±0.13**

TABLE V

COMPARISON ACCURACY (%) OF SF-MSFORMER WITH/WITHOUT MULTISCALE REPRESENTATION MODULE

Mode	Parameters	GFlops	OA(%)
without FPN+FMSR	0.21M	4.38	93.52
with FPN	11.82M	6.72	96.81
with FMSR	14.61M	17.21	97.94
with FPN+FMSR	26.12M	19.56	98.57

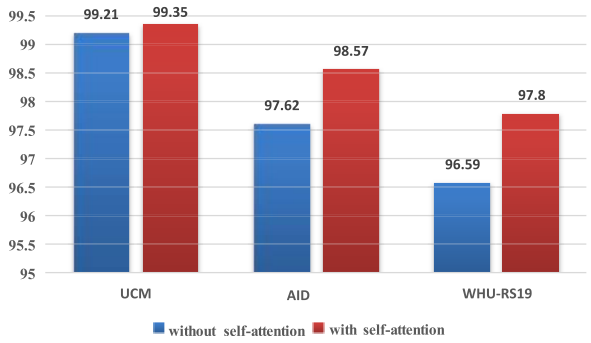


Fig. 14. Compare precision (%) of the SF-MSFormer with or without the self-attention module.

average accuracy improvement on the NWPU-RESISC45 dataset than ResNet18. The multiscale methods, such as EMTCAL, SCViT, MS2AP, and MF^{2}CNet, show better performance than the other methods. Among them, SCViT achieves the best performance in the contrasting methods. Compared with SCViT, the SF-MSFormer achieves competitive OA.

C. Ablations

The proposed SF-MSFormer consists of two multiscale branches. Thus, the ablation about the importance of the two branches and multiscale representation have been experimented. In addition, the proposed texture-enhanced encoder and self-attention module have been ablated in this section.

1) Effect of Different Multiscale Transformer Branch: The spatial multiscale Transformer branch (A Branch) and the frequency multiscale Transformer branch (B Branch) are integrated into the proposed SF-MSFormer. Thus, the performance of these two branches is shown in Fig. 10. It suggests that the performance of the spatial multiscale Transformer branch is superior to the frequency multiscale Transformer branch. When 50% of the dataset is used for training, the spatial multiscale Transformer branch shows about 2.84%, 0.68%, and 12.84% increased accuracy on the UCM, AID, and WHU-RS19 datasets, respectively. Compared with the performance of SF-MSFormer, the A branch contributes the most to SF-MSFormer's performance. Besides, it suggests that two components are integrated to show better average precision on the three datasets.

2) Effect of the Multiscale Representation Module: The proposed SF-MSFormer is ablated with/without the spatial multiscale representation and FMSR to validate the effects on the multiscale representation. The performance, parameters, and the GFlops of SF-MSFormer with/without the two-domain multiscale modules are shown in Table V. It shows that the multiscale representation is of great significance in SF-MSFormer. The SF-MSFormer with spatial–frequency multiscale features shows more significant performance improvements than the spatial or frequency multiscale features,

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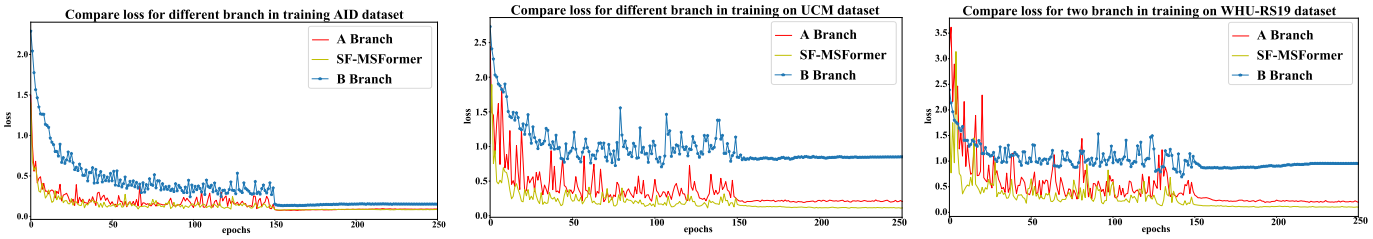


Fig. 15. Compare loss curve on (a) AID, (b) UCM, and (c) WHU-RS19 datasets (50% for testing), where A branch represents the spatial multiscale Transformer and B branch represents the proposed frequency multiscale Transformer.

TABLE VI
COMPARISON ACCURACY (%) OF THREE INPUT MODES FOR TEXTURE ENCODER

Input (Q,K,V)	WHU-RS19	AID	UCM
Q=K=V: LL	97.2	97.52	99.03
Q:LL, K=V:HL+LH+HH	97.8	98.57	99.15
Q:LL, K=V:LL+HL+LH+HH	**98.02**	**98.6**	**99.35**

though the parameters and the GFlops are increasing. Besides, the SF-MSFormer with spatial multiscale representation (FPN) has more remarkable performance improvement than with FMSR. The framework with FPN and FMSR can improve the performance in a coordinated manner. In addition, the detailed performances of these ablations about each category are shown in Fig. 11. It shows that multiscale representation performs well in the class with significant target size differences, such as airport, center, and stadium. For example, the OA of airport and the latter two categories is increased directly by 9% and 12%.

3) Effect of Different Input Modes on the Texture-Enhanced Encoder: To study the effect of different input modes on the texture-enhanced encoder, the ablations are designed with different input forms (shown as Table VI). In detail, LL mainly contains the abstract information of the image. When Q = K = V:LL, it shows only 97.2%, 97.52%, and 96.83% accuracy, respectively, on the WHU-RS19, AID, and UCM datasets. When K = V:HL + LH + HH or K = V:LL + HL + LH + HH, it shows improved accuracy than Q = K = V:LL. It shows that effectively combining the low-pass and high-pass components as the key and value of self-attention can obtain better performance for remote sensing scene classification. In contrast, K = V:LL + HL + LH + HH shows better performance than the other two input modes.

4) Effect of the Self-Attention Module: Self-attention plays an essential role in the proposed SF-MSFormer, which can introduce the global modeling ability to the multiscale representation. Thus, the ablation is experimented with by removing the two self-attention modules adopted in the SF-MSFormer.

In Fig. 12, the confusion matrices of SF-MSFormer with or without self-attention are shown. Fig. 12(a), (c), and (e) shows the confusion matrices of the SF-MSFormer without the self-attention module for three datasets. Fig. 12(b), (d), and (f) show the corresponding confusion matrices of the SF-MSFormer with the self-attention module on the three datasets above, respectively. It suggests that the number of categories with a recognition rate of 100% is increased from seven categories to 30 categories on the UCM dataset. Besides, it is clear that the confusion matrices are sparser, which shows better recognition performance of the proposed method with self-attention. In addition, the proposed method with the self-attention module shows quantitative performance improvements than those without the self-attention model as Fig. 13 shows. In detail, the proposed model with self-attention shows about 1.05%, 0.95%, and 1.21% increased accuracy than that without self-attention.

D. Visualization and Analysis

In this section, the visualization results are shown to validate the texture-enhanced encoder's explainability. Besides, the experimental results are analyzed and concluded.

1) Visualization: To show the SF-MSFormer's good representation ability, the feature visualizations of the proposed model are shown in Fig. 14. In detail, six examples of the UCM dataset are selected to show the proposed two-branch representation ability, including the baseball diamond, buildings, overpass, river, storage, and tennis court. The spatial multiscale Transformer (S_Cam) and the frequency multiscale Transformer branches' feature category activation map (F_Cam) are shown for each sample. It suggests that the texture-enhanced encoder focuses more on capturing the texture features than the spatial multiscale attention. Moreover, this also evaluates the effectiveness of the proposed texture-enhanced encoder.

2) Analysis: The experimental results show that the proposed SF-MSFormer performs better on the AID, UCM, and WHU-RS19 datasets and shows competitive performance on the NWPU-RESISC45 dataset, compared with the contrasting methods. In addition, the ablations offer that the spatial multiscale Transformer and frequency multiscale Transformer branches have different influences on the final recognition results. And the spatial multiscale Transformer branch dominates the performance for the final recognition precision. It is mainly because the frequency multiscale Transformer branch is more straightforward in the architecture design. Thus, learning the frequency multiscale features better can be a further direction to improve the performance of the SF-MSFormer framework. Besides, Fig. 12 suggests that the SF-MSFormer with the self-attention module can be distinguished for some categories. It shows that global feature is essential for the complex remote sensing scene.

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As for the computation complexity, the proposed SF-MSFormer shows competitive parameters and increased GFlops than most of the contrasting methods (shown in Tables II and V). Thus, how to optimize its computation complexity is still a further direction, especially in reducing the GFlops.

As for the convergency, the loss curves of these two branches and the final SF-MSFormer are shown in Fig. 15. It suggests that the final proposed SF-MSFormer is of faster and more stable convergency ability than those of A and B branches.

V. Conclusion

This article presents a multiscale spatial–frequency multiscale Transformer for remote sensing scene classification. It uses self-attention to model the global dependence of the spatial and frequency multiscale features, which can capture the global multiscale features better. In addition, it can enhance the representation ability, which is friendly to the recognition of the scenes with significant differences in size. The experiments demonstrate the effectiveness of the proposed SF-MSFormer, which shows better accuracy on the benchmark datasets, including AID, UCM, and WHU-RS19.

Besides, the SF-MSFormer is of strong feature representation ability. Thus, some directions are still worthy of further study, including the image segmentation task, object detection task, and so on. Besides, optimizing and reducing the transformer-based methods' computation complexity is still a further direction. In addition, there are many methods based on Transformer arising. Exploring more Transformer-based methods and making them adaptive to remote sensing scene classification will be our further work.

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